N-M-51G
July 1, 1975
SUPERSEDING
Int. Fed. Spec. N-M-0051F(Army-GL)
June 30, 1970 and
Fed. Spec. N-M-51E
June 24, 1967

FEDERAL SPECIFICATION

MACARONI, SPAGHETTI, VERMICELLI AND LASAGNA

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for preparation and packaging of macaroni, spaghetti, vermicelli, and lasagna for use by agencies of the Federal Government.

1.2 Classification. The products shall be of the following groups, types, classes, styles, and forms, as specified (see 6.1):

Groups:

I - Macaroni (all types)
II - Spaghetti (all types)
III - Vermicelli (all types)
IV - Lasagna (type A only)

Types:

A - Plain
B - Milk
C - Whole Wheat
D - Wheat and soy
E - Vegetable

Classes:

1 - Unenriched
2 - Enriched

Styles:

a - Regular cooking
b - Quick cooking

FSC 3920

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Forms:

i	-	Long
ii	-	Elbow (applicable to group 1 macaroni only)
iii	-	Twist (applicable to group 1 macaroni only)
iv	-	Rigatoni (applicable to group 1 macaroni only)
v	-	Shell (applicable to group 1 macaroni only)
vi	-	Curl (applicable to lasagna only)
vii	-	Flat (applicable to lasagna only)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

Federal Standards:

FED-STD-101	-	Preservation, Packaging and Packing Materials: Test Procedures
FED-STD-123	-	Marking for Domestic Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:

MIL-L-10547	-	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible
MIL-L-35078	-	Loads, Unit: Preparation of Nonperishable Subsistence

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-668	-	Sanitary Standards for Food Plants

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U. S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

Definitions and Standards for Macaroni and Noodle Products

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U. S. Department of Agriculture

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, U. S. Department Of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Post Office Box 540, Benjamin Franklin Station, Washington, DC 20044.)

American Association of Cereal Chemists

Approved Methods

(Copies may be obtained from the American Association of Cereal Chemists 1821 University Ave., St. Paul, MN 55104.)

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National Research Council

Food Chemicals Codex

(Application for copies should be addressed to the Office of Director, National Academy of Sciences, National Research Council, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials. The basic ingredients from which all types, classes, styles, and forms of macaroni, spaghetti, vermicelli, and lasagna are formed shall consist of semolina, durum wheat flour, farina, or hard wheat flour (other than durum wheat flour) or any combination of two or more of these ingredients and a moistening ingredient. For military agencies the farinaceous ingredient shall be a good quality semolina which will result in a product satisfying the requirements specified in table II and the carotenoid color test. The farinaceous ingredient or combination of ingredients may be specified in the invitation for bids or request for proposal (see 6.1 and 6.4). All supplementary ingredients i.e., vegetable solids, vitamin enrichment, type E or substitute ingredients in the case of type B are provided for in the requirements for the various types, classes, and styles of the product. Glyceryl monostearate conforming to definitions and standards for macaroni and noodle products, under the Federal Food, Drug, and Cosmetic Act, may be included as an optional ingredient; however, glyceryl monostearate shall be included when specified in the invitation for bids or request for proposal (see 6.1). All materials used shall be edible, clean, wholesome, of good quality and free from evidence of insect infestation and other objectionable matter and foreign flavors and odors.

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3.2 Types.

3.2.1 Type A, plain. This type shall consist of dried formed units of dough made of ingredients described in 3.1, or combinations thereof, and water. Gum gluten may be used as an optional ingredient in such amounts that the protein content of the farinaceous ingredient and gum gluten combined shall not exceed 13 percent of the weight of the finished product.

3.2.2 Type B milk. This type shall be made of the farinaceous ingredients specified in 3.2.1 except that milk instead of water shall be used as the sole moistening ingredient in preparing the dough; or in lieu of milk, concentrated milk, evaporated milk, dry whole milk, a mixture of butter and skim milk, concentrated skim milk, nonfat dry milk solids (dried skim milk) with or without water, or any two or more of these may be used. The proportion of nonfat dry milk solids and total milk solids used in such mixture shall conform to Definitions and Standards for Macaroni, and Noodle Products under Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

3.2.3 Type C, whole wheat. This type shall consist of dried units of dough made from whole hard wheat flour or whole durum wheat flour or both, and water.

3.2.4 Type D, wheat and soy. This type shall be made of the ingredients specified in 3.2.1, except that soy flour shall be added in a quantity not less than 12.5 nor more than 15.0 percent of the combined weight of the wheat and soy ingredients used. The soy flour used shall be made from heat-processed, dehulled soybeans with or without removal of fat therefrom.

3.2.5 Type E, vegetable. This type shall be made of the ingredients specified in 3.2.1, except that tomato (of any red variety), artichoke, beet, carrot, parsley, or spinach, as specified, is added in such quantity that the solids thereof is not less than three nor more than five percent by weight of the finished product.

3.3 Classes.

TABLE I. Enrichment requirements

Ingredients	Per pound basis	
	Minimum	Maximum
Required:		
Thiamine, mg.	4.0	5.0
Riboflavin, mg.	1.7	2.2
Niacin or niacinamide, mg.	27.0	34.0
Iron, mg.	13.0	16.5
Optional:		
Calcium, mg.	500	625
Partly defatted wheat germ, percent	---	5.0

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3.3.1 Class 1, unenriched, macaroni, spaghetti, vermicelli, and lasagna. The unenriched product shall meet all requirements for its particular type, as specified in 3.2.1 through 3.2.5.

3.3.2 Class 2, enriched macaroni, spaghetti, vermicelli, and lasagna shall comply with requirements as specified in 3.3.1 and table I. In addition to all the required ingredients, class 2 products may contain any of the optional ingredients specified in table I within the minimum and maximum amount. Only those optional ingredients specified in invitation for bids or request for proposal may be used (see 6.1).

3.3.2.1 The thiamine, riboflavin, niacin, or niacinamide, iron, and calcium may be supplied wholly or in part by the use of dried yeast, partly defatted wheat germ, enriched farina, or enriched flour or by the direct addition of the enrichment ingredients.

3.3.2.2 Iron and calcium may be added only in forms that are harmless and assimilable. Thiamine, riboflavin, niacin, or niacinamide and iron may be added in a harmless carrier which does not impair the enriched product, such carrier being used only in the quantity reasonably necessary to effect an intimate and uniform distribution of such substances in the finished enriched product.

3.4 Styles.

3.4.1 Style a, regular cooking macaroni, spaghetti, vermicelli, and lasagna. Regular cooking macaroni, spaghetti, vermicelli, and lasagna shall conform with all requirements for the product, as specified in 3.3.1 or 3.3.2.

3.4.2 Style b, quick cooking macaroni, spaghetti, vermicelli, and lasagna. Quick cooking macaroni, spaghetti, vermicelli, and lasagna shall conform with requirements as specified in 3.4.1. In addition, style b products shall contain not less than 0.5 percent and not more than 1.0 percent of disodium phosphate ($Na_2HPO_47H_2O$). The disodium phosphate shall conform to the requirements of the Food Chemicals Codex.

3.5 Form.

3.5.1 Macaroni. Macaroni of all types shall be in the form of tube shaped units, reasonably uniform in diameter, i.e., more than 0.11 inches (2.8 mm) but not more than 0.27 inch (6.9 mm) in diameter except for forms iii, iv, and v.

3.5.1.1 Form i, long macaroni. The unit of long macaroni shall be essentially straight and not less than 8-1/2 (21.6 cm) nor more than 11 inches (27.9 cm) in length. The outside diameter shall be not less than 0.110 inches (2.8 mm) and not more than .230 inches (5.8 mm) and the thickness shall be not less than .030 inches (0.8 mm) and not more than .040 inches (1.0 mm).

3.5.1.2 Form ii, elbow macaroni. The unit of elbow macaroni shall be elbow-shaped and not less than 0.75 (19.1 mm) nor more than 1.5 inches (3.8 cm) in outer curvilinear length. The outside diameter shall be not less than .160 inches (4.1 mm) and not more than .200 inches (5.1 mm) and the thickness shall be not less than .030 inches (0.8 mm) and not more than .038 inches (1.0 mm).

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3.5.1.3 Form iii, twist (Gemelli), macaroni. The unit of twist macaroni shall be not less than 1.25 (3.2 cm) nor more than 1.5 inches (3.8 cm) in length or .03 ± .001 inches (0.8 ± .03 mm) in thickness.

3.5.1.4 Form iv, rigatoni macaroni. The unit of rigatoni macaroni shall be tubular shaped with ribbed surface and not less than 1.25 (3.2 cm) nor more than 1.5 inches (3.8 cm) in length. The diameter shall be not less than 0.59 (15.0 mm) nor more than 0.68 inch (17.3 mm), and the thickness shall be not less than .036 (0.9 mm) nor more than .044 inch (1.1 mm).

3.5.1.5 Form v, shell macaroni. The unit shell macaroni shall be sea shell shaped with following sizes and dimensions:

	LENGTH (inches)	THICKNESS (inch crest to inside wall)
Large size	minimum 1-1/4 (3.2 cm) - maximum 1-3/4 (4.4 cm)	.040 (1.0 mm) - .045 (1.1 mm)
Medium size	minimum 3/4 (19.0 mm) - maximum 1-1/8 (2.86 cm)	.035 (0.9 mm) - .040 (1.0 mm)
Small size	minimum 7/16 (11.1 mm) - maximum 5/8 (15.9 mm)	.031 (0.8 mm) - .036 (0.9 mm)

3.5.2 Spaghetti. Spaghetti of all types shall be in the form of tube-shaped or solid cylindrical units, reasonably uniform in diameter, i.e., more than .060 inches (1.5 mm) but not more than .083 inches (2.1 mm) in diameter. The units shall be not less than 8-1/2 inches (21.6 cm) nor more than 11 inches (27.9 cm) in length. When specified (see 6.1), spaghetti shall be more than 0.06 (1.5 mm) but not more than 0.08 inch (2.0 mm) in diameter.

3.5.3 Vermicelli. Vermicelli of all types shall be in the form of solid cylindrical units, reasonably uniform in diameter, and more than 0.40 inches (1.0 mm) but not more than 0.59 inch (1.5 mm) in diameter. The units shall be essentially straight and not less than 8-1/2 inches (21.6 cm) nor more than 11 inches (27.9 cm) in length.

3.5.4 Lasagna. Lasagna of all types shall be in the form of solid sheets essentially flat and straight in form.

3.5.4.1 Form vi. The unit of curl lasagna shall be 2-1/2 (6.4 cm)) + 1/8 (3.2 mm) or - 1/2 inch (12.7 mm) wide and 9-1/2 (24.1 cm) ± 1/2 inch (12.7 mm) in length and shall be .040 to .054 inches (1.0 to 1.4 mm), in thickness. The two long edges shall be ruffled throughout the length of the unit.

3.5.4.2 Form vii. The unit of flat lasagna shall be 1-1/2 (3.8 cm) ± 1/8 inch (3.2 mm) wide and 9-1/2 (24.1 cm) ± 1/4 inch (6.4 mm) in length and shall be .040 to .054 inches (1.0 to 1.4 mm) in thickness. The edges shall be cut straight.

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3.6 Chemical requirements. Macaroni, spaghetti, vermicelli, and lasagna shall conform to the requirements shown in table II.

TABLE II. Chemical requirements

Type	Minimum limit of protein 1/ (N x 5.7)	Minimum Carotenoid Color 2/	Maximum limits of Ash 1/	Maximum limits of Moisture
	Percent	p.p.m	Percent	Percent
A	12.2	2.5	0.90 2/ 4/ .60 3/ 4/	13.0
B	12.7	2.5	.98 2/ .80 3/	13.0
C	13.2	2.5	2.5	13.0
D	15.2	2.5	1.7	13.0
E	12.1	2.5	1.5	13.0

1/ Calculated to a moisture-free basis.

2/ Applicable to products in which the principal farinaceous ingredients are semolina or durum wheat flour or a combination thereof.

3/ Applicable to products in which the principal farinaceous ingredients are farina or hard wheat flour other than durum wheat flour or a combination thereof.

4/ Exclusive of the ash contributed by disodium phosphate in all quick-cooking products, and by the added calcium in enriched macaroni, spaghetti, vermicelli, and lasagna, in which calcium has been used as an optional ingredient.

3.7 Workmanship. The product shall be prepared in accordance with part 128 of the Code of Federal Regulations, Title 21, Human Foods; Current Good Manufacturing Practice (sanitation) in manufacture, processing, packing or holding. In addition, for Military agencies the product shall be prepared in plants meeting the requirements of MIL-STD-668.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preacceptance inspection (plant sanitation inspection).

4.2.1 For civil agencies. When specified (see 6.1), inspection of plants producing the product will be conducted upon the request of the contracting officer to the appropriate Government inspection agency. Award of contracts will be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval recommended by the inspection agency.

4.2.1.1 For military agencies. The product shall be rejected if produced in plants not meeting the requirements of 3.7.

4.3 Acceptance inspection.

4.3.1 Condition inspection of containers. When inspection of product for quality and condition is performed by the U. S. Department of Agriculture, the United States Standards for Condition of Food Containers shall apply. (For applicable criteria governing inspection of containers for the Military, see 4.3.4.1).

4.3.2 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.3 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.3.1 Testing of box liners or bag. Box liners and bags shall be tested to determine conformance to 5.1.1.1.1. Test procedures for basis weight of paper, weight of wax or wax-polyethylene, thickness of polyethylene and melting point of wax shall be as referenced in 5.1.1.1.1. The sample unit for testing shall be one square yard or equivalent in sheets. The sample size shall be five sample units. Lot size shall be expressed in sheets. Test requirements shall be average requirements. Each test result shall be reported to the same unit of measurement as specified for the pertinent requirement. A test failure shall indicate unacceptable liners, and use of such shall be cause for rejection of the lot.

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4.3.4 Process examination. Examination shall be performed during manufacture of the product to determine conformance to 3.1, 3.2 and 3.3, as concerns kinds, quality and condition of ingredients. Examination shall consist of organoleptic evaluations, as necessary, and review of pertinent labels, invoices or other valid documents. Where specific limitations are imposed on certain ingredients or additives, formulation records shall be maintained. Nonconformance shall be cause for rejection of the involved finished product.

4.3.4.1 Examination of end item. Examinations shall be in accordance with tables IV through VII and 4.3.4.2 through 4.3.4.4. Sampling plans for the aforementioned tables shall be in accordance with table III. The acceptable quality levels (AQLs) shall be expressed as defects per hundred units for table IV, VI and VII, and percent defective for table V.

TABLE III. Sampling plans

Table	Sample unit	Lot size expressed in	Inspection level	AQLs Major	AQLs Minor
IV	One filled and closed primary container	Primary containers	I	1.5	4.0
V	One filled box	Primary containers	S-3	---	2.5
VI	One filled and closed primary container	Primary containers	S-2	1.5	4.0
VII	1/4 pound (113.5 g) from a primary container	Primary containers	S-2	1.5	6.5

TABLE IV. Examination of primary containers

Categories		Defect
Major	Minor	
101		Containers not type or size specified
102		Tear, hole, or open closure or seam
103		Not closed in the manner specified (see 5.1.1.1.2, or 5.1.2.1 as applicable)

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TABLE IV. Examination of primary containers (cont'd)

Categories		Defect
Major	Minor	
104		Box closure insecure 1/
105		Not clean
106		Bag or wrap has odor
107		Nomenclature or cooking directions missing, incorrect or illegible
	201	Other required label information missing, incorrect or illegible

1/ Applicable to taped closure specified in 5.1.1.1.2. A defect shall be scored if less than 75 percent of the tape secures flaps.

TABLE V. Examination for net weight 1/

Category	Defect
Minor	
201	More than five percent under specified weight (containers to have a net weight of one pound (0.45 kg) or less)
202	More than two percent under specified weight (containers required to have a net weight of more than one pound (0.45 kg) but not more than 10 pounds (4.5 kg))
203	More than one percent under specified weight (containers required to have a net weight of more than 10 pounds (4.5 kg))

1/ Report results to nearest 1/4 ounce (7. g) for container required to have one pound (0.45 kg) or less; nearest 1/2 ounce (14. g) for container required to have more than one pound (0.45 kg) but not more than 10 pounds (4.5 kg); nearest ounce for containers required to have more than 10 pounds (4.5 kg). Lot shall be rejected if sample data indicates a lot average net weight less than specified net weight.

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TABLE VI. Examination of liners or bags when required

Categories		Defect
Major	Minor	
101		Missing
102		Not type specified
	201	Liners not "U" shaped in container for spaghetti, macaroni, vermicelli, or lasagna
103		Not completely protecting product from container
104		Not clean
	202	Polyethylene bag not closed with a twist tie

TABLE VII. Examination of product characteristics 1/

Categories		Defect
Major	Minor	Examination of uncooked product
	201	Shape not as specified
101		Color not typical of the product
102		Objectionable odor
103		Moldy or otherwise off condition
	202	More than 25 percent by count of product not within limitation for length 2/
104		Diameter, thickness or width of product not within limits specified 3/
	203	More than 10 percent by weight but not more than 25 percent by weight of product has black specks, checks, or has misshapen or broken units; or combinations thereof
105		More than 25 percent by weight of the product has black specks, checks, or has misshapen or broken units; or combination thereof

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TABLE VII. Examination of product characteristics 1/ (cont'd)

Categories		Defect
Major	Minor	Examination of cooked product 4/
106		Does not possess a characteristic flavor or odor
107		Not free from undesirable flavors or odors (e.g., rancid, bitter, musty, etc.)
108		Not firm and tender

1/ Presence of foreign material (e.g., filth, insects, glass, metal, dirt, paint, etc.) shall be cause for rejection of the lot.

2/ Measure twenty pieces (except measure one piece of lasagna) from a sample unit. The pieces shall not be broken or misshapen. Use a linear measuring device calibrated in 1/16 inch (1.6 mm) increments, or a suitable go-no-go device.

3/ Measure one piece from a sample unit. The piece shall not be broken or misshapen. Use a caliper or micrometer calibrated in .001 inches (.03 mm) for all products except lasagna. For lasagna use a measuring device calibrated in 1/64 inch (0.4 mm) increments. A suitable go-no-go gauge may be substituted for any measuring device.

4/ See 4.4.1.1 for procedure.

4.3.4.2 Examination of shipping container. Examination of the filled and closed shipping containers shall be in accordance with the quality assurance provisions in the appendix of PPP-B-636. In addition, the following defects shall be included in the examination: Major - (1) Markings missing, incorrect or illegible; (2) - Protective pad and tape when required, missing, not as specified, or not completely covering metal stitches; (3) - Intermediate corrugated fiberboard box when required, missing; Minor - (1) Number of interior packages not as required; (2) Not snug-fitting, for level C pack as specified in 5.2.3.1 only the defects pertinent to marking and number of interior packages shall apply.

4.3.4.3 Examination of case liners. The filled and closed waterproof case liners shall be examined in accordance with the quality assurance provisions in the appendix of MIL-L-10547.

4.3.4.4 Examination of unit loads. Unit loads shall be examined in accordance with the quality assurance provisions of MIL-L-35078.

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4.3.5 Testing of finished product. The finished product shall be tested for the characteristics specified in table II as qualified therein. In addition, when class 2 product is procured, testing shall be performed to determine conformance to table I. The sample for testing shall be a one-pound (0.45 kg) composite of product derived from the number of containers indicated by inspection level S-2. Lot size shall be expressed in terms of the primary container. Each test result shall be reported to the same unit or decimal as specified for the pertinent requirement. A test failure shall be cause for rejection of the lot.

4.4 Examination and test procedures. Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum such quality assurance, the supplier will use the procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

4.4.1 Examination procedures.

4.4.1.1 Cooking. The product shall be cooked in accordance with the manufacturer's instruction. The cooked product shall be tender, not mushy, firm, possess a characteristic flavor and odor, and shall be free from undesirable flavors and odors.

4.4.2 Test procedures.

4.4.2.1 Chemical tests. All chemical and other tests required to determine the protein, ash, moisture, and enrichment ingredient content shall be made in accordance with the Official Methods of Analysis of the Association of Offical Analytical Chemists in effect on the date of invitation for bids, or request for proposal or in accordance with methods that give equivalent results.

4.4.2.2 Carotenoid color of macaroni, spaghetti, vermicelli, and lasagna. The minimum carotenoid pigment color score (see table II) of the packaged product shall be determined in accordance with method 14-50 in American Association of Cereal Chemists, Approved Methods.

4.4.2.3 Filth tests. When specified (see 6.1), all test required to determine compliance with the provisions of the Federal Food, Drug, and Cosmetic Act, in regard to insects, insect fragments, rodent hairs, and rodent excreta shall be made in accordance with the methods of the Association of Official Analytical Chemists, in effect on the date of invitation for bids or request for proposal, or in accordance with methods that give equivalent results.

4.4.3 Percentages. All percentages shall be upon the basis of weight.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level B or C, as specified (see 6.1).

5.1.1 Level B, military agencies.

5.1.1.1 Macaroni (forms i and ii), spaghetti, vermicelli, and lasagna. The product, of the group specified (see 6.1), shall be packaged in the quantity and size of unit container indicated in table VIII. The container shall be a fiberboard box constructed in conformance with style RSC, class domestic of PPP-B-636. The box shall be lined and closed as specified in 5.1.1.1.1 and 5.1.1.1.2, respectively.

5.1.1.1.1 Liners or bags. Two U-shaped strips of waxed paper shall be placed in the box to separate the product from the top, bottom, sides and ends of the box. The base paper shall have a commercial basis weight of 20 pounds (9.1 kg) per ream (24 by 36-500) when tested in accordance with method 5022 of FED-STD-101, and shall be wet-waxed on both sides with a total of not less than eight pounds (3.6 kg) per ream of fully refined paraffin wax having a melting point of not less than 130 °F (54.4 °C) when tested in accordance with methods 6018 and 4017, respectively, of FED-STD-101. Alternatively, the sheet may be overwaxed with not less than 12 pounds (5.4 kg) per ream of a coating composed of a blend of polyethylene and wax. Test method for determining weight of alternate coating shall be the same as that for wax coating. Alternatively, the product shall be enclosed in a 0.003-inch (.08 mm) thick polyethylene bag. The bag shall be closed with a twist tie. Thickness of the polyethylene shall be determined in accordance with L-P-378.

5.1.1.1.2 The top and bottom flaps of containers shall be sealed with tape conforming to PPP-T-45, type III, grade B, minimum width 2 inches (50.8 mm). The tape shall be applied the full length of the joint formed by the closure of the outer flaps and extend onto the end panels a minimum of 2 inches (50.8 mm).

5.1.1.2 Macaroni (all forms), spaghetti, vermicelli, and lasagna (both forms). When specified (see 6.1), one pound (0.45 kg) of the product shall be packaged in a carton normally used by the manufacturer of the product. Ten, twelve, twenty, or twenty-four, one-pound (0.45 kg) cartons shall be packed in an intermediate fiberboard box normally used by the manufacturer as a domestic shipping container for those quantities of the product.

5.1.2 Level B, civil agencies.

5.1.2.1 One and five pound quantities. One (1) pound (0.45 kg) or five (5) pounds (2.3 kg) of the product, of one description only, shall be packaged in accordance with the manufacturer's standard practice.

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5.1.2.2 Ten or twenty pound quantities. Ten (10) pound (4.5 kg) or twenty (20) pound (9.1 kg) quantities shall be packaged in a plastic bag of a type approved for food use. The bag shall be of appropriate size and design with closure accomplished by heat-sealing or by use of a suitable tape.

5.1.3 Level C, all agencies. The product shall be packaged in a manner to protect it against deterioration during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. The product in the quantities indicated in tables VIII or IX, as applicable, shall be packed in a snug-fitting shipping container in accordance with level A, B or C, as specified (see 6.1).

5.2.1 Level A. The shipping container shall be a snug-fitting fiberboard box, constructed, closed, and reinforced in accordance with style RSC, V2s of PPP-B-636. The shipping container shall be provided with a waterproof case liner, fabricated and sealed in accordance with MIL-L-10547 for subsistence items. When flaps of the shipping container are closed by stitching, the case liner shall be protected from the stitching with a paperboard pad placed between the stitches and the case liner. The paperboard pad shall have the same dimensions as the shipping container with a minus tolerance of 3/16 inch (4.8 mm). The case liner shall be protected from stitches in the manufacturer's joint by a piece of pressure-sensitive tape placed over the stitches.

5.2.2 Level B. The shipping container shall be a snug-fitting fiberboard box, constructed and closed in accordance with style RSC, V3c, V3s or V4s of PPP-B-636. The shipping container shall be provided with a waterproof case liner and protection from stitching in accordance with 5.2.1.

5.2.2.1 When specified (see 6.1 and 6.2), the shipping container specified in 5.2.2 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636, except that two reinforcing bands shall be used, one lengthwise and one girthwise.

5.2.3 Level C. The shipping container shall be a snug-fitting fiberboard box, constructed and closed in accordance with style RSC, type CF or SF, class domestic, method II closure of PPP-B-636.

5.2.3.1 When specified (see 6.1 and 6.3), the shipping container shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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TABLE VIII. Quantities and container sizes for Military agencies

Groups	Unit package quantities and container sizes	Shipping container quantities
Macaroni, long and elbow	9 lbs. (4.1 kg) - 11-1/4" (28.6 cm) by 8-5/8" (21.9 cm) by 4-3/4" (12.1 cm)	4 (36 lbs.) (16.3 kg)
Spaghetti and vermicelli	10 lbs. (4.5 kg) - 11-1/4" (28.6 cm) by 8-5/8" (21.9 cm) by 4-3/4" (12.1 cm)	4 (40 lbs.) (18.1 kg)
Lasagna	10 lbs. (4.5 kg) - 11-1/4" (28.6 cm) by 8-5/8" (21.9 cm) by 4-3/4" (12.1 cm)	4 (40 lbs.) (18.1 kg)
Group I - Forms ii, iii, iv and v		
Macaroni, elbow twist and rigatoni, shells	1 lb. (0.45 kg) 1 lb. (0.45 kg) 5 lbs. (2.3 kg) 10 lbs. (4.5 kg)	10 (10 lbs.) (4.5 kg) 20 (20 lbs.) (9.1 kg) 2 (10 lbs.) (4.5 kg) 1 (10 lbs.) (4.5 kg)
Groups I, II and III - Form i		
Macaroni, long spaghetti and vermicelli	1 lb. (0.45 kg) 1 lb. (0.45 kg) 1 lb. (0.45 kg) 5 lbs. (2.3 kg) 10 lbs. (4.5 kg) 20 lbs. (9.1 kg)	12 (12 lbs.) (5.4 kg) 20 (20 lbs.) (9.1 kg) 24 (24 lbs.) (10.9 kg) 4 (20 lbs.) (9.1 kg) 2 (20 lbs.) (9.1 kg) 1 (20 lbs.) (9.1 kg)
Group IV		
Lasagna	1 lb. (0.45 kg) 10 lbs. (4.5 kg)	12 (12 lbs.) (5.4 kg) 2 (20 lbs.) (9.1 kg)

5.3 Unit loads. When specified (see 6.1) the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.4 Labeling and marking.

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5.4.1 Unit containers. Any labeling that complies with the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder. In addition, cooking directions shall be included in the label.

5.4.2 Fiberboard box overpacks and shipping containers. Fiberboard box overpacks and shipping containers shall be marked in accordance with FED-STD-123 or MIL-STD-129, as applicable.

5.4.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Group, type, class, style, or form required (see 1.2).
(c) Kind(s) of farinaceous ingredients required (see 3.1).
(d) When glyceryl monostearate is required (see 3.1).
(e) Optional enrichment, if any (see 3.3).
(f) When special spaghetti diameter size is required (see 3.5.2).
(g) When plant sanitation inspection is required by Civil agencies (see 4.2.1).
(h) When filth test is required (see 4.4.2.3).
(i) Level of packaging required (see 5.1).
(j) When one-pound cartons are required for Military agencies (see 5.1.1.2).
(k) Quantities required for level B packaging for Civil agencies (see 5.1.2.1 and 5.1.2.2).
(l) Level of packing required (see 5.2).
(m) When packing specified in 5.2.2.1 (see 6.2) or 5.2.3.1 (see 6.3) is required.
(n) Type and class of unit load when unit loading is specified (see 5.3).

6.2 Packing specified in 5.2.2.1 is intended for transfer at sea operations or specified overseas operations.

6.3 Packing specified in 5.2.3.1 is intended for direct shipment from the supply source to the first consuming activity within CONUS for immediate use.

6.4 Farinaceous ingredient. Semolina is generally considered to be the preferred farinaceous ingredient for the production of macaroni products of the highest acceptability.

6.5 Destination inspection (for civil agencies). When the finished product has been inspected and passed at point other than destination, further inspection for quality should not be necessary; however, the contract may require that the product be inspected at destination for condition and quantity.

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6.6 Military agencies. Based on conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.14D/DSAR 4145.7.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Preparing activity:

Army - GL

Project No. 8920-0287

Review activities:

Army - MD
Navy - MC, MS
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☆U.S. GOVERNMENT PRINTING OFFICE: 1976-703-020/1013

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